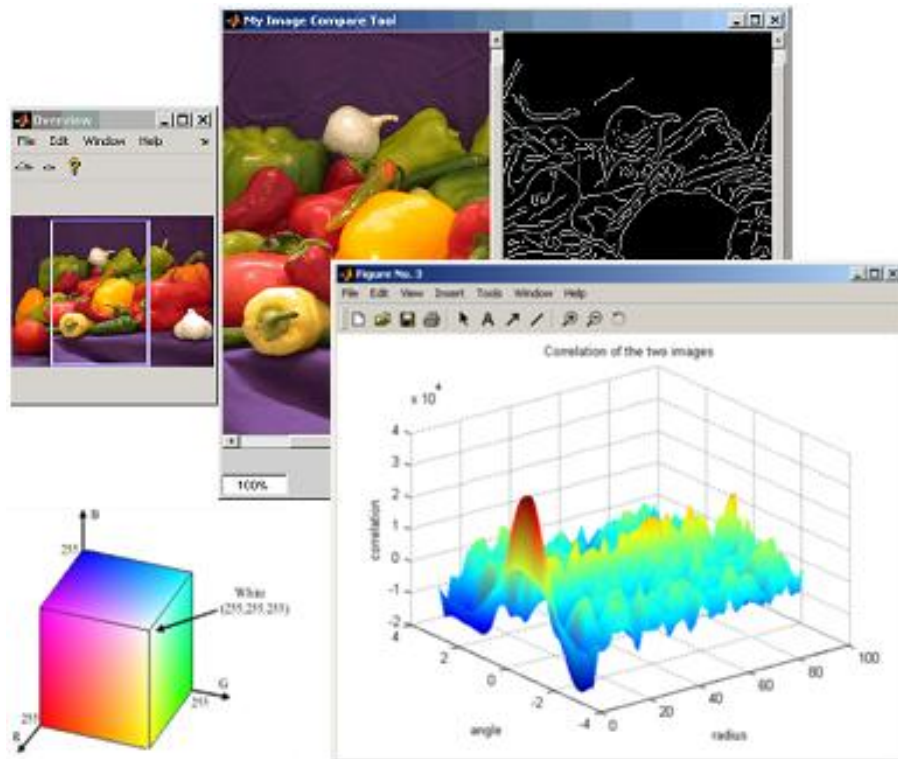


Digital Image Processing

LECTURE 12: FILTERING IN FREQUENCY DOMAIN

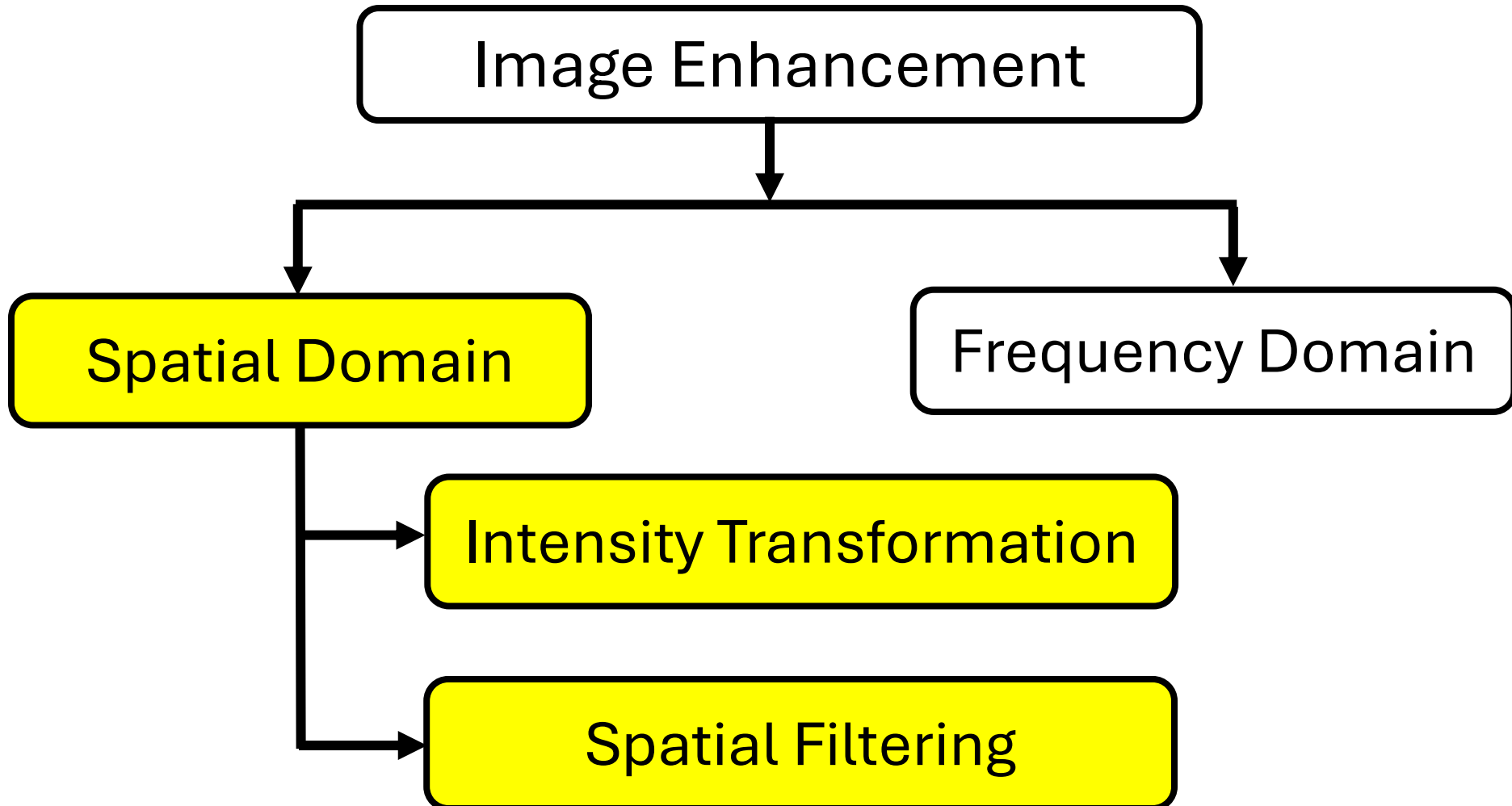


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GLA University Mathura

Image Enhancement (Frequency Domain)

Image Enhancement

❑ Classification of Image Enhancement Techniques:



Fourier Series

Fourier Series

The Fourier series is a mathematical tool **used to represent periodic functions** as an **infinite sum of sine and cosine functions**.

Fourier Series

- A periodic function $f(x)$ with a period T can be expressed as a Fourier series:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos \left(\frac{2\pi nx}{T} \right) + b_n \sin \left(\frac{2\pi nx}{T} \right) \right]$$

Where

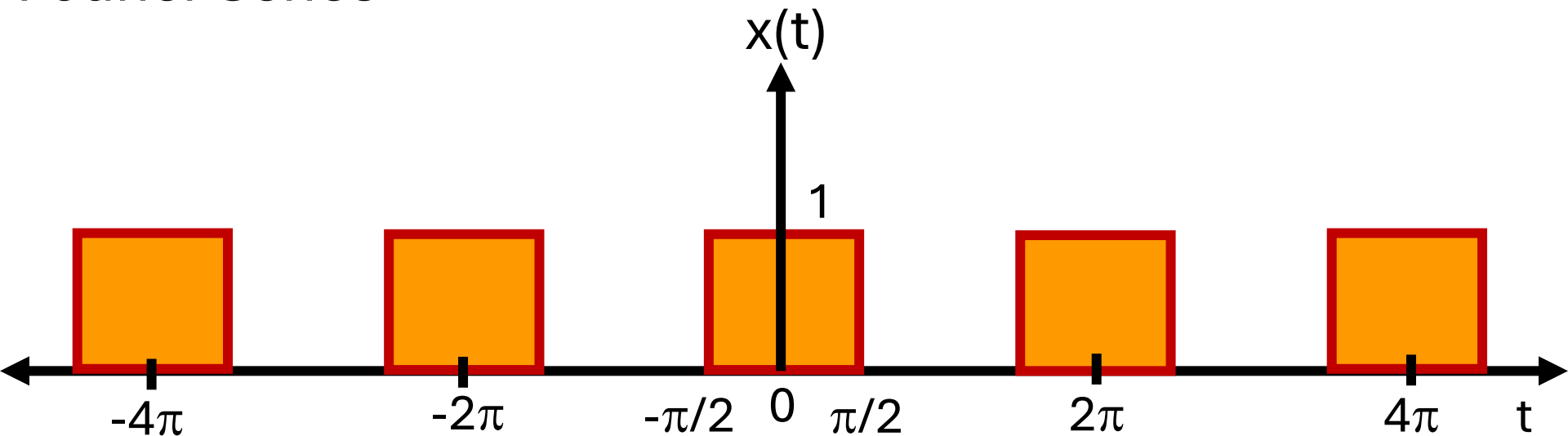
$$a_0 = \frac{1}{T} \int_0^T f(x) dx$$

$$a_n = \frac{2}{T} \int_0^T f(x) \cos \left(\frac{2\pi nx}{T} \right) dx$$

$$b_n = \frac{2}{T} \int_0^T f(x) \sin \left(\frac{2\pi nx}{T} \right) dx$$

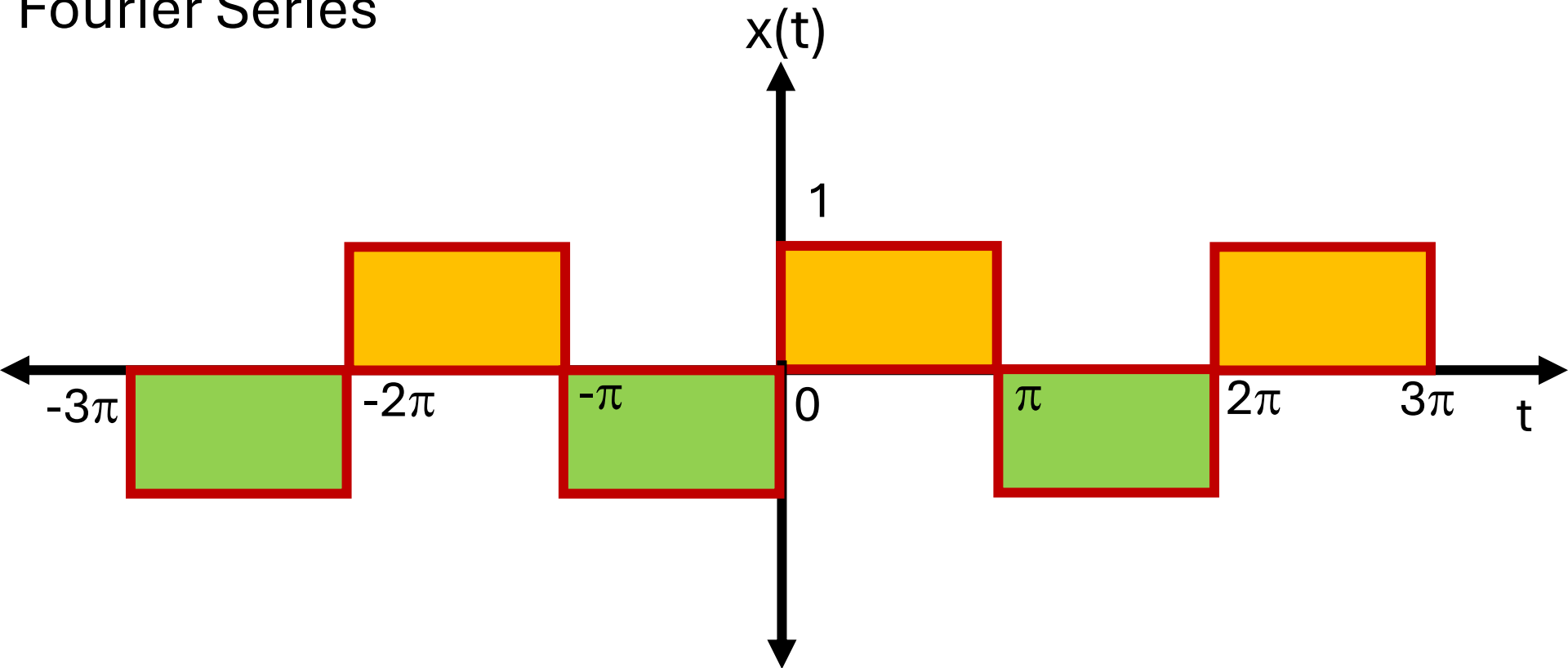
Numerical Example

Ques1: For the following waveform $x(t)$, compute the Fourier Series



Numerical Example

Ques 2: For the following waveform $x(t)$, compute the Fourier Series



Fourier Transform

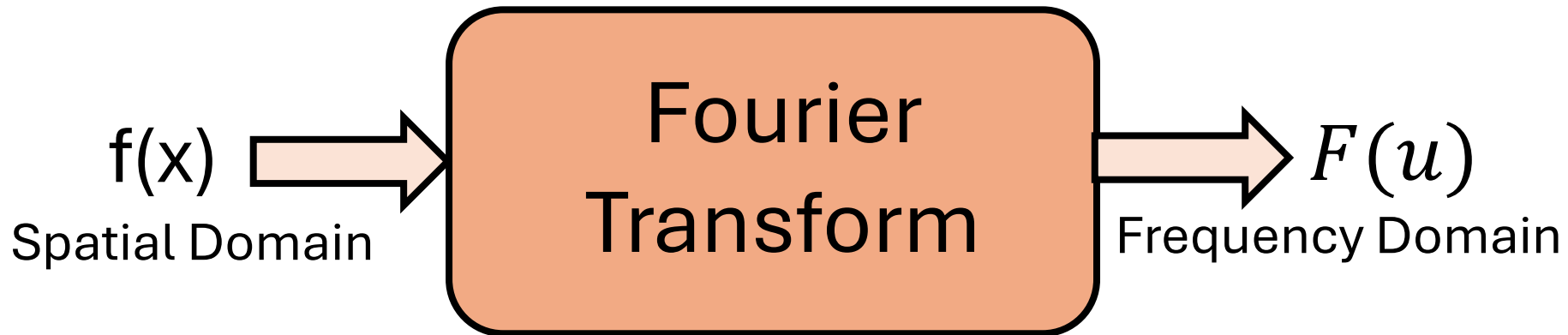
Fourier Transformation

- The Fourier Transform is a powerful mathematical technique that transforms **a function of time, denoted as $x(t)$, into a function of frequency**, represented as $X(\omega)$.
- This transformation allows for the **analysis of signals in terms of their frequency components**, providing a framework to understand and manipulate various types of data in fields such as signal processing, physics, and engineering.

Fourier Transformation

- The Fourier transform is obtained from the **Fourier Integral in Complex Form.**

Filtering in Frequency Domain



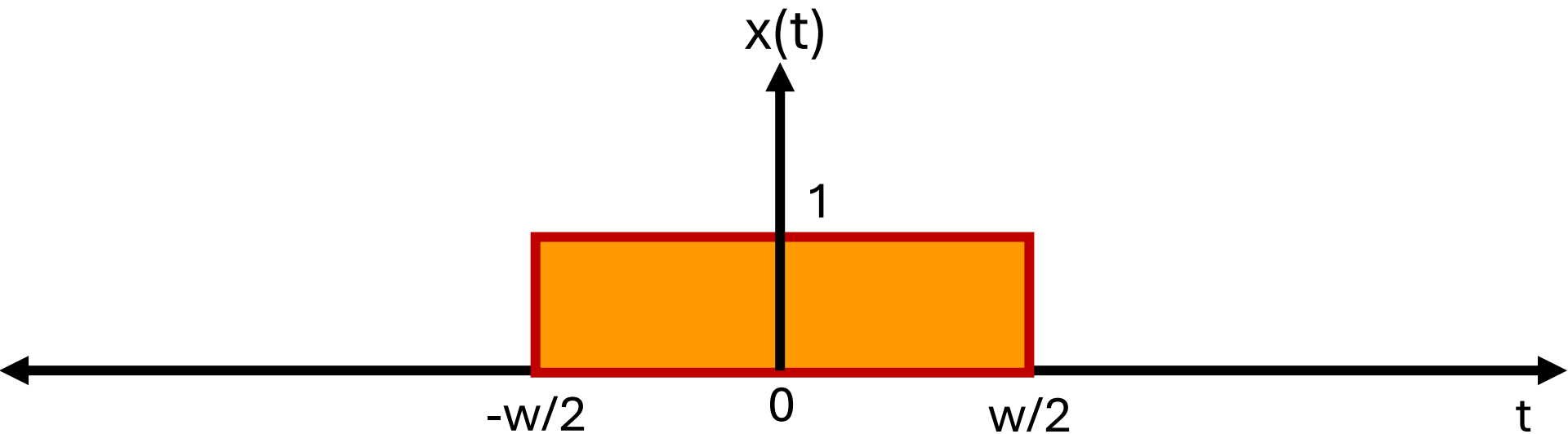
Continuous Fourier Transformation

□ 1-D Continuous Fourier Transformation:

$$\mathfrak{F}(f(x)) = F(u) = \int_{-\infty}^{\infty} f(x) e^{-j2\pi ux} dx$$

Numerical Example

Ques 2: Compute the Fourier Transformation for the following waveform:



Continuous Fourier Transformation

□ $F(u)$ and $f(x)$ are Fourier Transform Pair.

□ $F(u)$ is a complex function:

$$F(u) = R(u) + j \cdot I(u) = |F(u)|e^{j\phi(u)}$$

□ Fourier Spectrum:

$$|F(u)| = \sqrt{(R(u))^2 + (I(u))^2}$$

□ Phase angle:

$$\phi(u) = \tan^{-1} \left(\frac{I(u)}{R(u)} \right)$$

□ Power Spectrum:

$$P(u) = |F(u)|^2 = (R(u))^2 + (I(u))^2$$

Continuous Fourier Transformation

□ 2-D Continuous Fourier Transformation:

$$F(u, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy$$

Numerical Example

Ques3: For the following two-dimensional function, perform the Fourier Transformation:

$$f(x, y) = \begin{cases} 1 & \text{for } -\frac{X}{2} \leq x \leq \frac{X}{2}, -\frac{Y}{2} \leq y \leq \frac{Y}{2} \\ 0 & \text{otherwise} \end{cases}$$

Continuous Fourier Transformation

□ $F(u, v)$ and $f(x, y)$ are Fourier Transform Pair.

□ $F(u, v)$ is a complex function:

$$F(u, v) = R(u, v) + j \cdot I(u, v) = |F(u, v)| e^{j\phi(u, v)}$$

□ Fourier Spectrum:

$$|F(u, v)| = \sqrt{(R(u, v))^2 + (I(u, v))^2}$$

□ Phase angle:

$$\phi(u, v) = \tan^{-1} \left(\frac{I(u, v)}{R(u, v)} \right)$$

□ Power Spectrum:

$$P(u, v) = |F(u, v)|^2 = (R(u, v))^2 + (I(u, v))^2$$

Continuous Fourier Transformation

□ 1-D Continuous Inverse Fourier Transformation:

$$\mathfrak{F}^{-1}(F(u)) = f(x) = \int_{-\infty}^{\infty} F(u) e^{j2\pi ux} du$$

Continuous Fourier Transformation

□ **2-D Continuous Inverse Fourier Transformation:**

$$f(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv$$

Discrete Fourier Transformation

1-D Discrete Fourier Transformation

□ Mathematical Equation of One Dimensional (1-D) Discrete Fourier Transformation (DFT)

The Fourier transform of a discrete function of one-variable $f(x)$, for $x = 0, 1, 2, \dots, M-1$ is given by the following equations:

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j \frac{2\pi u x}{M}}$$

for $x = 0, 1, 2, \dots, M-1$

Numerical Example on 1-D DFT

Ques1: Find the Discrete Fourier Transformation (DFT) of the $f(x) = \{0, 1, 2, 1\}$

Ans:

For $f(x) = \{0, 1, 2, 1\}$, $M = 4$

Substituting $M = 4$ in the Mathematical Equation of 1-D DFT,

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j \frac{2\pi ux}{M}}$$

Substituting $M = 4$ in the Mathematical Equation of 1-D DFT,

$$F(u) = \sum_{x=0}^3 f(x) e^{-j \frac{\pi ux}{2}}$$

Numerical Example on 1-D DFT

F (u = 0)

$$= \sum_{x=0}^3 f(x) e^{-j \frac{\pi \cdot 0 \cdot x}{2}} = \sum_{x=0}^3 f(x) e^{-j0}$$

$$= f(0) + f(1) + f(2) + f(3) = 0 + 1 + 2 + 1 = \mathbf{4}$$

F (u = 1)

$$= \sum_{x=0}^3 f(x) e^{-j \frac{\pi \cdot 1 \cdot x}{2}} = \sum_{x=0}^3 f(x) e^{-j \frac{\pi x}{2}}$$

$$= f(0) \cdot e^{-j0} + f(1) \cdot e^{-j \frac{\pi}{2}} + f(2) \cdot e^{-j\pi} + f(3) \cdot e^{-j \frac{3\pi}{2}}$$

$$= 0 + 1(-j) + 2(-1) + 1(-j)(-1) = \mathbf{-2}$$

F(u = 2)

$$= \sum_{x=0}^3 f(x) e^{-j \frac{\pi \cdot 2 \cdot x}{2}} = \sum_{x=0}^3 f(x) e^{-j\pi x}$$

$$= f(0) e^{-j0} + f(1) e^{-j\pi} + f(2) e^{-j2\pi} + f(3) e^{-j3\pi}$$

$$= 0 + 1(-1) + 2(1) + 1(-1) = \mathbf{0}$$

Numerical Example on 1-D DFT

$$= \sum_{x=0}^3 f(x) e^{-j \frac{\pi \cdot 3 \cdot x}{2}} = \sum_{x=0}^3 f(x) e^{-j \frac{3\pi x}{2}}$$

F (u = 3)

$$\begin{aligned} &= f(0) + f(1)e^{-j \frac{3\pi}{2}} + f(2)e^{-j 3\pi} + f(3)e^{-j \frac{9\pi}{2}} \\ &= 0 + 1(-j)(-1) + 2(-1) + 1(-j)(1) = -2 \end{aligned}$$

$$F(u) = \{4, -2, 0, -2\}$$

Numerical Example on 1-D DFT

❑ Alternate Approach to determine 1-D DFT:

Twiddle Matrix (Kernel)
for 1-D DFT

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

Formula for 1-D DFT using Twiddle Matrix: $= [\text{Kernel}] \times f(x)$

Numerical Example on 1-D DFT

Ques: Perform the DFT for the following $f(x) = \{0, 1, 2, 1\}$

$$\begin{aligned}
 \begin{bmatrix} F(u=0) \\ F(u=1) \\ F(u=2) \\ F(u=3) \end{bmatrix} &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} f(x=0) \\ f(x=1) \\ f(x=2) \\ f(x=3) \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 + 1 + 2 + 1 \\ 0 - j - 2 + j \\ 0 - 1 + 2 + 1 \\ 0 + j - 2 - j \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \\ 0 \\ -2 \end{bmatrix}
 \end{aligned}$$

$$F(u) = \{4, -2, 0, -2\}$$

1-D Inverse Discrete Fourier Transformation

□ Mathematical Equation of One-Dimensional (1-D) Inverse Discrete Fourier Transformation (IDFT)

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j \frac{2\pi ux}{M}}$$

for $u = 0, 1, 2, \dots, M-1$

Numerical Question on 1-D IDFT

Ques2: Find the IDFT of the following $F(u) = \{4, -2, 0, -2\}$

Ans: for $F(u) = \{4, -2, 0, -2\}$,

$M = 4$; Substituting the Value of M ,

$$\begin{aligned} f(x) &= \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j \frac{2\pi ux}{M}} &= \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \frac{2\pi ux}{4}} \\ & &= \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \frac{\pi ux}{2}} \end{aligned}$$

Numerical Question on 1-D IDFT

$f(x = 0)$

$$\begin{aligned}
 &= \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \frac{\pi u \cdot 0}{2}} = \frac{1}{4} \sum_{u=0}^3 F(u) e^{j0} \\
 &= \frac{1}{4} (F(0) + F(1) + F(2) + F(3)) = \frac{1}{4} (4 - 2 + 0 - 2) \\
 &= \mathbf{0}
 \end{aligned}$$

$f(x = 1)$

$$\begin{aligned}
 &= \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \frac{\pi u \cdot 1}{2}} = \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \frac{\pi u}{2}} \\
 &= \frac{1}{4} \left[F(0) e^{j0} + F(1) e^{j \frac{\pi}{2}} + F(2) e^{j\pi} + F(3) e^{j \frac{3\pi}{2}} \right] \\
 &= \frac{1}{4} [4 + (-2)(j) + 0 + (-2)(-j)] = \mathbf{1}
 \end{aligned}$$

Numerical Question on 1-D IDFT

$$\begin{aligned}
 f(x=2) &= \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \frac{\pi u \cdot 2}{2}} = \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \pi u} \\
 &= \frac{1}{4} [F(0)e^{j0} + F(1)e^{j\pi} + F(2)e^{j2\pi} + F(3)e^{j3\pi}] \\
 &= \frac{1}{4} [4 + (-2)(-1) + 0 + (-2)(-1)] = \mathbf{2}
 \end{aligned}$$

$$\begin{aligned}
 f(x=3) &= \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \frac{\pi u \cdot 3}{2}} = \frac{1}{4} \sum_{u=0}^3 F(u) e^{j \frac{3\pi u}{2}} \\
 &= \frac{1}{4} [F(0)e^{j0} + F(1)e^{j\frac{3\pi}{2}} + F(2)e^{j2\pi} + F(3)e^{j\frac{9\pi}{2}}] \\
 &= \frac{1}{4} [4 + (-2)(-j) + 0 + (-2)(j)] = \mathbf{1}
 \end{aligned}$$

$$f(x) = \{0, 1, 2, 1\}$$

Numerical Question on 1-D IDFT

❑ Alternate Approach to compute One-Dimensional (1-D) IDFT:

$$\text{Twiddle Matrix (Kernel) for 1-D IDFT} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

Formula for 1-D DFT using Twiddle Matrix: $= \frac{1}{M} [\text{Kernel}] \times F(u)$

Numerical Question on 1-D IDFT

Ques: Perform the IDFT of the following $F(u) = \{4, -2, 0, -2\}$

$$\begin{bmatrix} f(x=0) \\ f(x=1) \\ f(x=2) \\ f(x=3) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} F(u=0) \\ F(u=1) \\ F(u=2) \\ F(u=3) \end{bmatrix}$$

$$= \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 0 \\ -2 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 4 - 2 + 0 - 2 \\ 4 - 2j + 0 + 2j \\ 4 + 2 + 0 + 2 \\ 4 + 2j + 0 - 2j \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \\ 1 \end{bmatrix}$$

$$f(x) = \{0, 1, 2, 1\}$$

2-D Discrete Fourier Transformation

□ Mathematical Equation of Two-Dimensional (2-D) Discrete Fourier Transformation (DFT):

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

for $x = 0, 1, 2, \dots, M-1$ and $y = 0, 1, 2, \dots, N-1$

2-D Discrete Fourier Transformation

❑ **Alternate Approach for 2-D Discrete Fourier Transform:**

Twiddle Matrix (Kernel) for 2-D DFT =
$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

Formula for 2-D DFT using Twiddle Matrix:
$$= [Kernel] \times f(x, y) \times [Kernel]^T$$

2-D Discrete Fourier Transformation

Ques3: Perform DFT for the following

$$f(x, y) = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} F(u, v) &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix} \end{aligned}$$

2-D Inverse Discrete Fourier Transformation

□ Mathematical Equation of Two-Dimensional (2-D) Inverse Discrete Fourier Transformation (IDFT):

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

for $u = 0, 1, 2, \dots, M-1$ and $v = 0, 1, 2, \dots, N-1$

2-D Inverse Discrete Fourier Transformation

❑ Alternate Approach for 2-D IDFT:

Twiddle Matrix (Kernel) for 2-D IDFT =
$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}$$

Formula for 2-D IDFT using Twiddle Matrix

$$= \left(\frac{1}{MN} \right) [Kernal] \times F(u, v) \times [Kernel]^T$$

Numerical Example on 2-D DFT

Ques3: Perform 2-D IDFT for the following

$$F(u, v) = \begin{bmatrix} 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix}$$

$$\begin{aligned} f(x, y) &= \frac{1}{16} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \\ &= \frac{1}{16} \begin{bmatrix} 8 & 8 & 8 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Basic Steps of Filtering in Frequency Domain

Basic Steps in Frequency Domain

1. Calculate the DFT of the image f
2. Generate a frequency domain filter H
 - H and F should have the same size
 - H should NOT be centered. Centered H is for displaying purpose only.
 - If H is centered, F needs to be centered too, and some post-processing is required
3. Multiply F by H (element by element)
4. Compute IDFT
5. Take the real part of the IDFT

Basic Steps in Frequency Domain

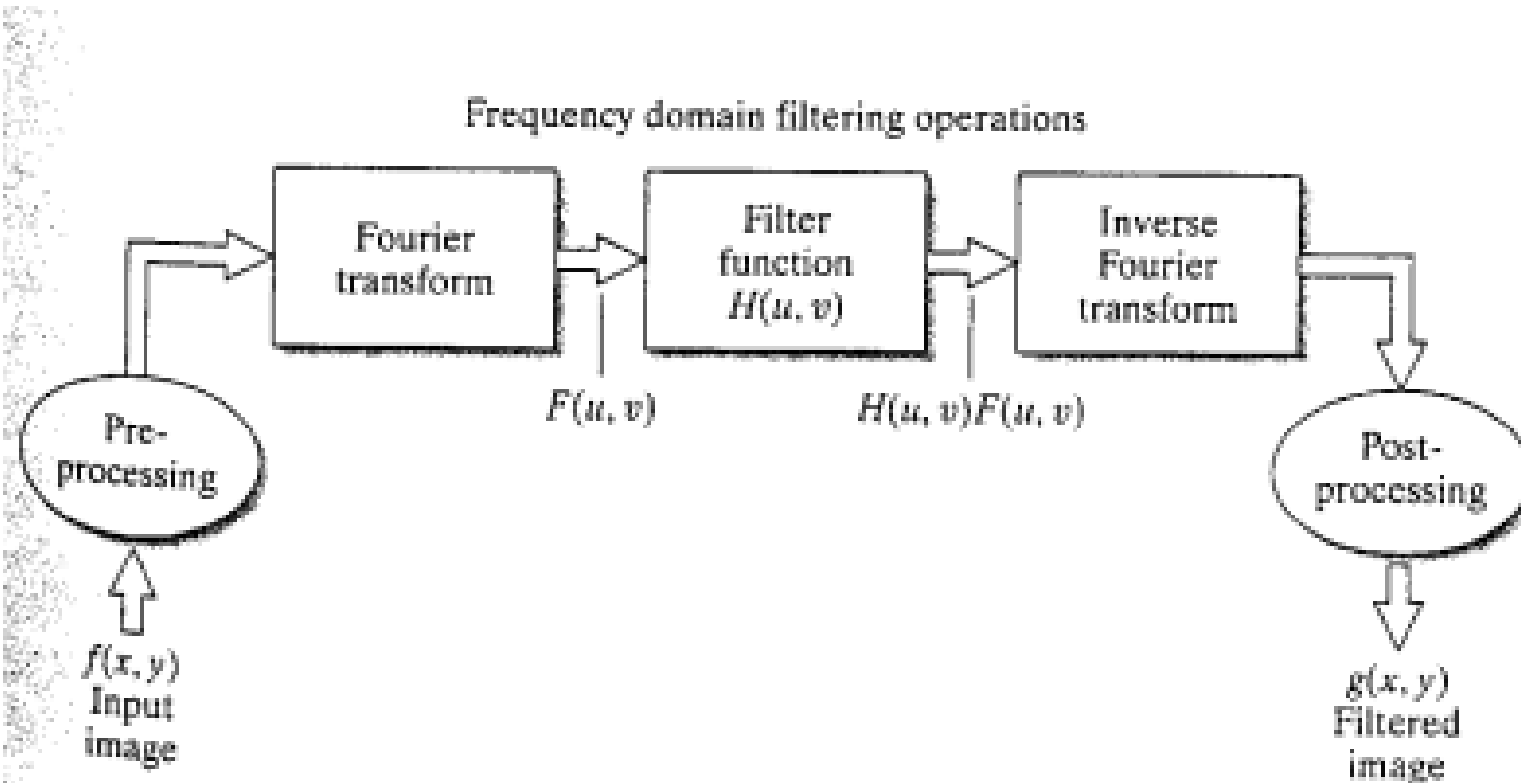
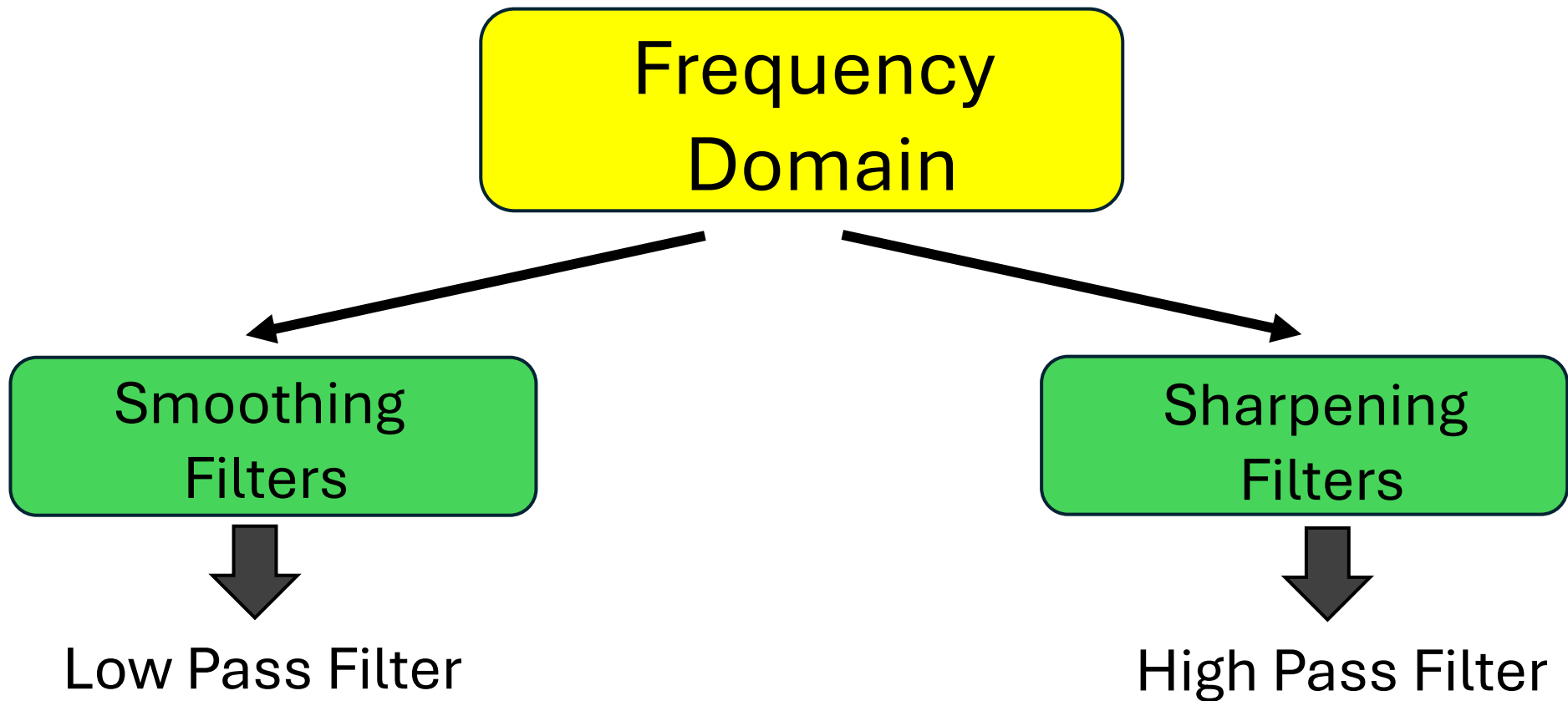


FIGURE 4.8

Basic steps for filtering in the frequency domain.

Operations Performed in Frequency Domain



Low Pass Filter Vs High Pass Filter

- ❑ **Low Pass Filter:** Lowpass filtered image have fewer sharp details than the original image
 - Attenuates high frequencies
 - Passes low frequencies

- ❑ **High Pass Filter:** High pass filtered image have less gray level variations in the smooth areas & emphasized transitional (e.g. edge) gray level details. Such an image appears sharper.
 - Attenuates low frequencies
 - Passes high frequencies

Low Pass Filter Vs High Pass Filter

❑ Low Pass Filter:

- Low pass filter are used for image smoothing in the frequency domain
- It removes high frequency noise from the image and preserves low frequency components

❑ High Pass Filter:

- High pass filter are used for image sharpening in the frequency domain
- It removes low frequency noise from the image and preserves high frequency components

Low Pass Filters

Low Pass Filter

- Removes Noise
- Blur the image
- Noise usually contribute in high-frequency domain/component
- So, to stop noise (high frequency component), we use low pass filter and thus, smoothen the image.

❑ Types of Low Pass Filters:

1. Ideal Low Pass Filter (ILPF)
2. Butterworth Low Pass Filter (BLPF)
3. Gaussian Low Pass Filter (GLPF)

Low Pass Filter

❑ Ideal Low Pass Filter (ILPF):

- Sharp Cut off

❑ Butterworth Low Pass Filter (BLPF):

- Depending on the order of the filter, it offers very sharp or smooth transition.
- High order BLPF, behaves like Ideal Low Pass Filter
- Low order BLPF, behaves like Gaussian Low Pass Filter.

❑ Gaussian Low Pass Filter (GLPF):

- Smooth Transition

Ideal Low Pass Filter

□ Mathematical Equation of Ideal Low Pass Filter:

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

Where,

D_0 is a positive constant,

$D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the $P \times Q$ frequency rectangle; that is

$$D(u, v) = \left[\left(u - P/2 \right)^2 + \left(v - Q/2 \right)^2 \right]^{1/2}$$

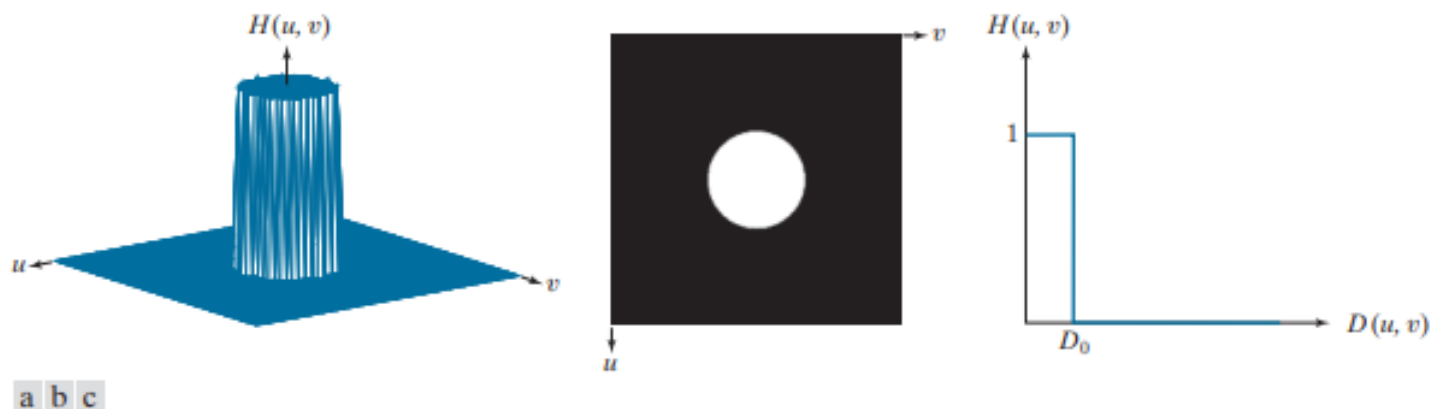


FIGURE 4.39 (a) Perspective plot of an ideal lowpass-filter transfer function. (b) Function displayed as an image. (c) Radial cross section.

Ideal Low Pass Filter

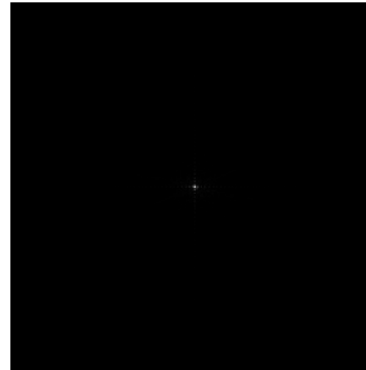
Original Image f



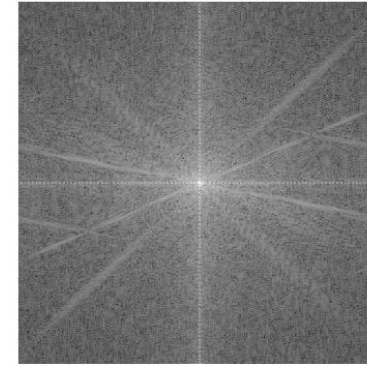
Fourier Spectrum F



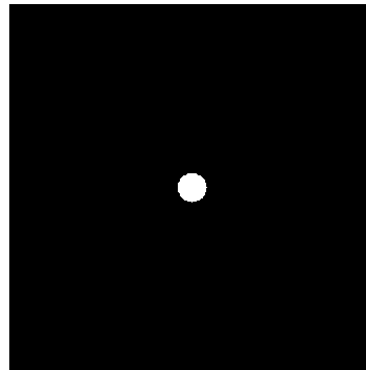
Centered Fourier Spectrum F_c



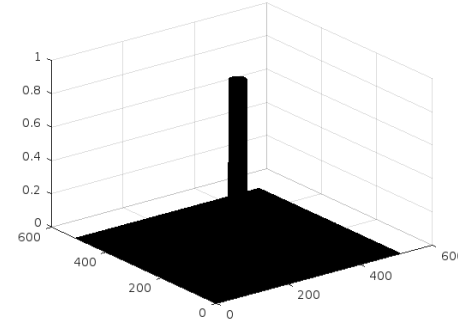
S_2 : Log Transformed F_c



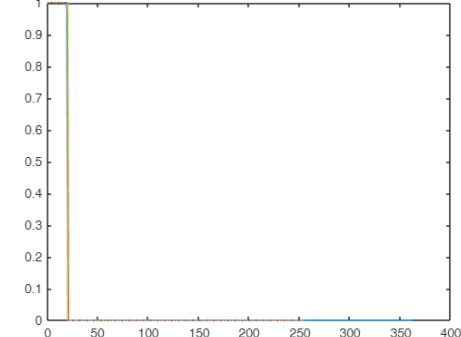
2D view of H of Ideal LPF



3D View of H of Ideal LPF

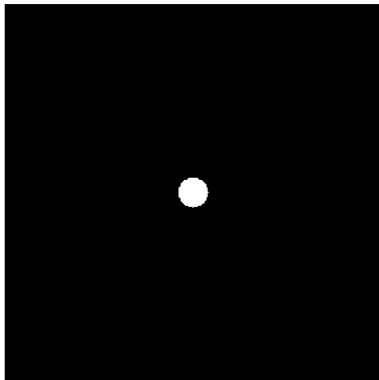


Transfer Function of H of Ideal LPF



Ideal Low Pass Filter

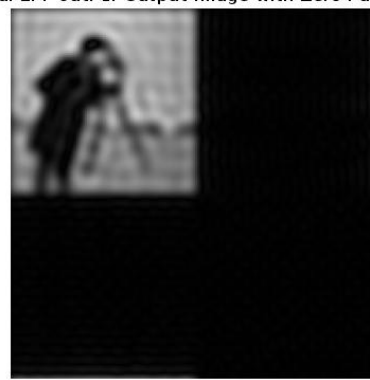
Ideal LPF Output Image: out



Ideal LPF Centered Output Image: outc



Ideal LPF outF1: Output Image with Zero Padding



Ideal LPF outF2: Output Image without zero padding



Gaussian Low Pass Filter

□ Mathematical Equation of Gaussian Low Pass Filter:

$$H(u, v) = e^{-D^2(u, v) / 2(D_0)^2}$$

Where,

D_0 : be the Cut-off frequency.

$D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the $P \times Q$ frequency rectangle; that is

$$D(u, v) = \left[\left(u - P/2 \right)^2 + \left(v - Q/2 \right)^2 \right]^{1/2}$$

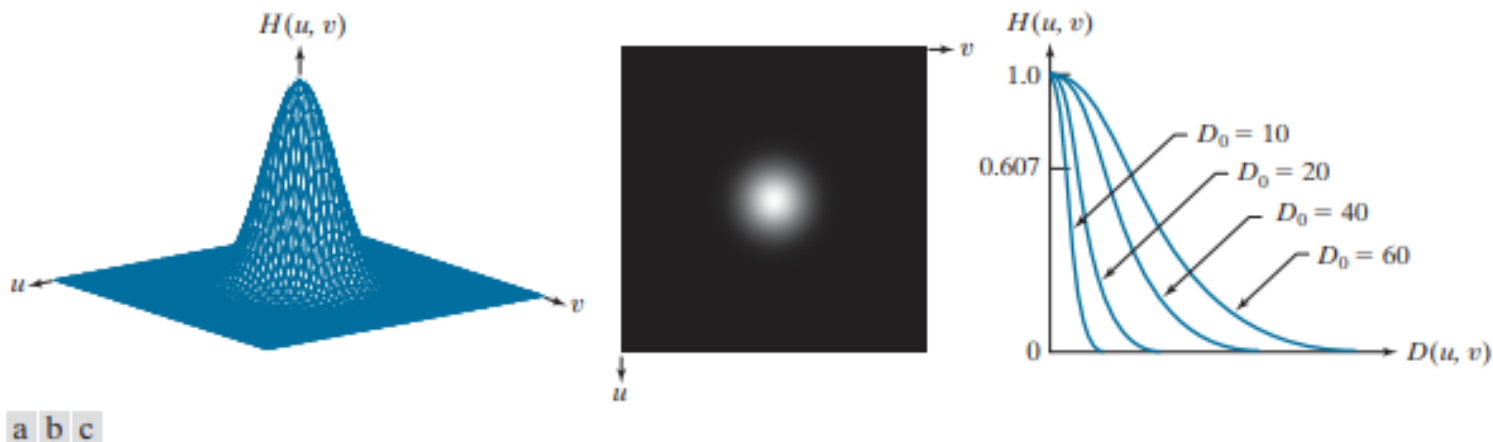


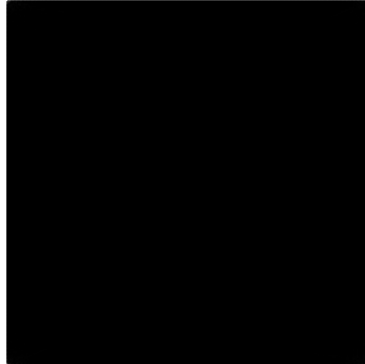
FIGURE 4.43 (a) Perspective plot of a GLPF transfer function. (b) Function displayed as an image. (c) Radial cross sections for various values of D_0 .

Gaussian Low Pass Filter

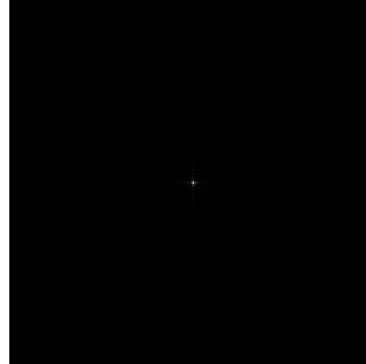
Original Image f



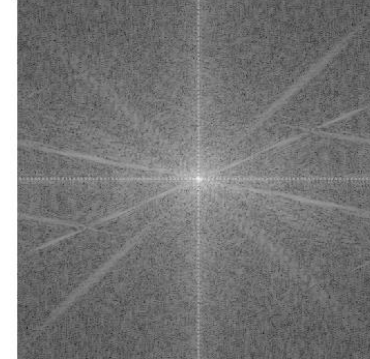
Fourier Spectrum F



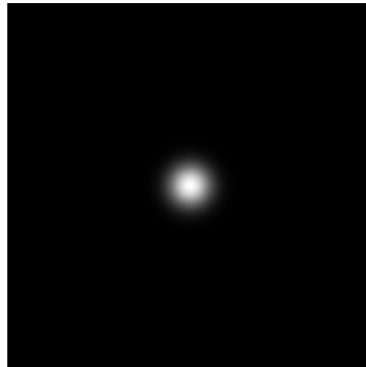
Centered Fourier Spectrum F_c



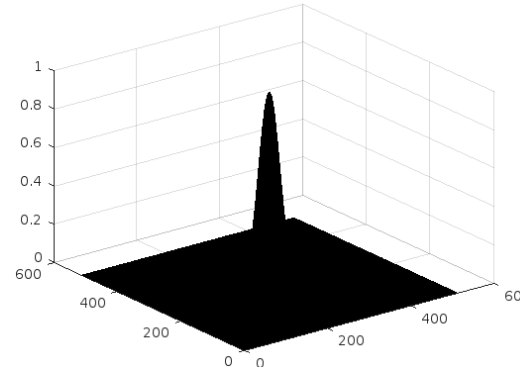
S_2 : log transformed F_c



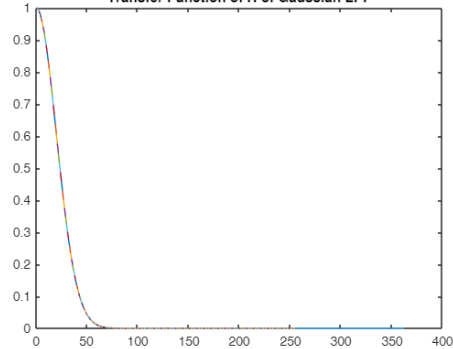
2D View of H of Gaussian LPF



3D View of H of Gaussian LPF

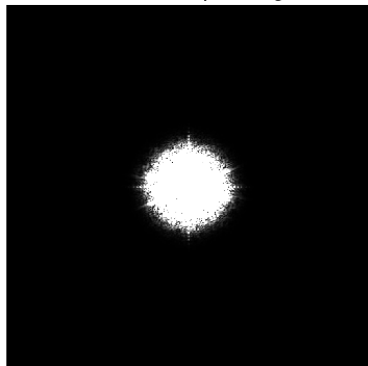


Transfer Function of H of Gaussian LPF

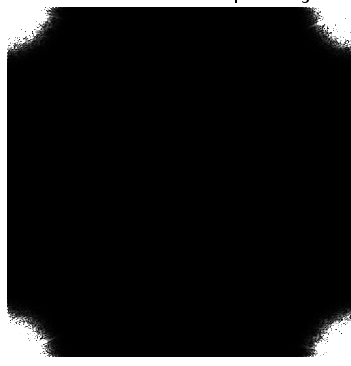


Gaussian Low Pass Filter

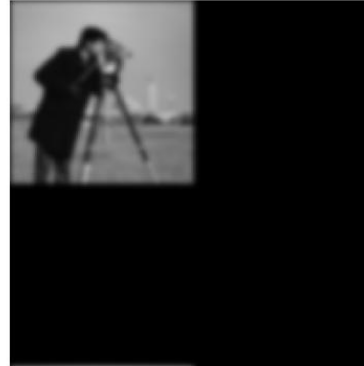
Gaussian LPF Output Image:out



Gaussian LPF Centered Output Image:outc



Gaussian LPF outF1: Output Image with Zero Padding



Gaussian LPF outF2: Output Image without Zero Padding



Butterworth Low Pass Filter

Mathematical Equation of Butterworth Low Pass Filter:

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)}{D_0} \right]^{2n}}$$

Where,

n : be the order of the filter and

$D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the $P \times Q$ frequency rectangle; that is

$$D(u, v) = \left[\left(u - P/2 \right)^2 + \left(v - Q/2 \right)^2 \right]^{1/2}$$

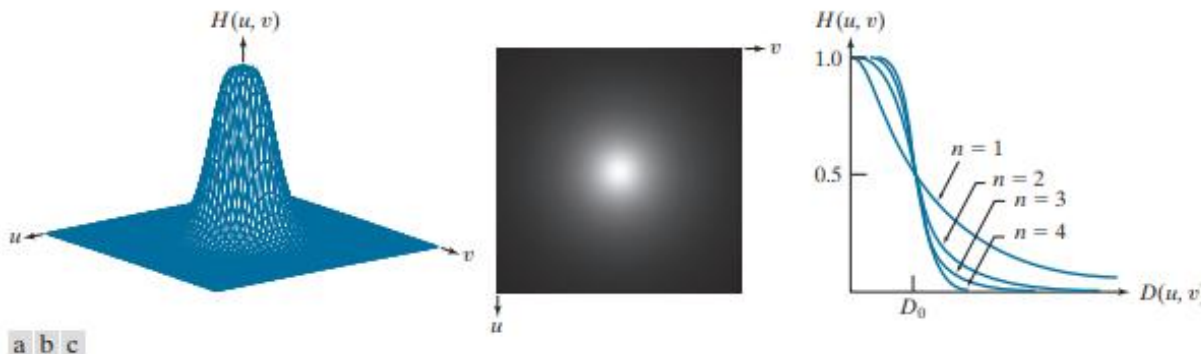


FIGURE 4.45 (a) Perspective plot of a Butterworth lowpass-filter transfer function. (b) Function displayed as an image. (c) Radial cross sections of BLPFs of orders 1 through 4.

Butterworth Low Pass Filter

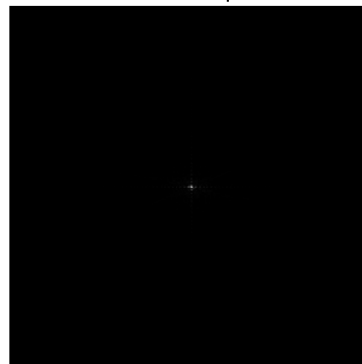
Original Image f



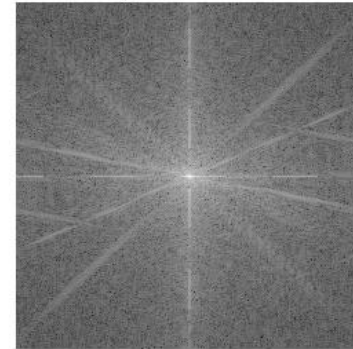
Fourier Spectrum F



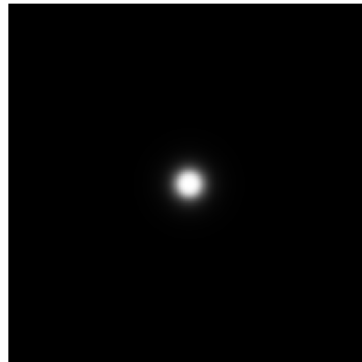
Centered Fourier Spectrum F_c



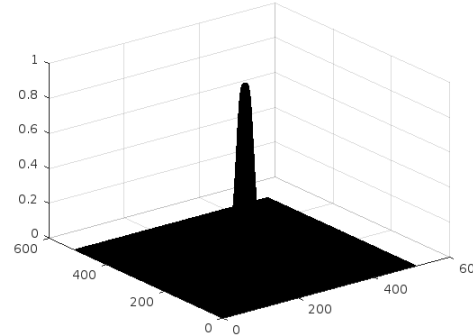
S_2 : Log Transformed F_c



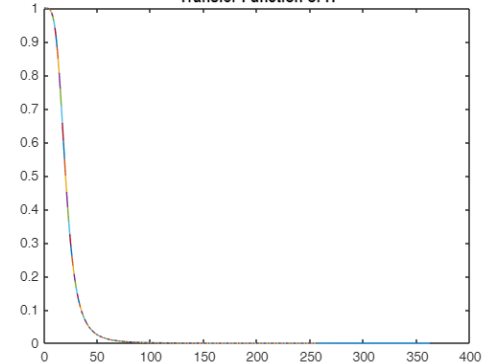
2D View of Butterworth LPF



3D view of H

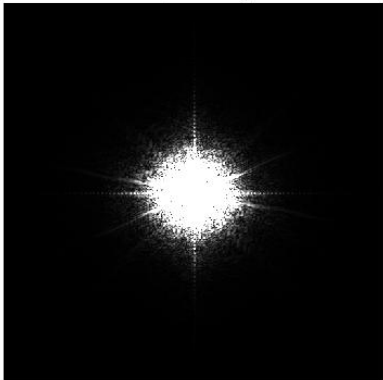


Transfer Function of H

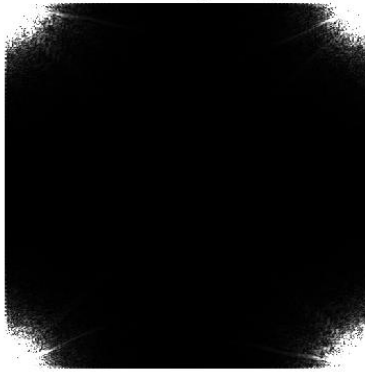


Butterworth Low Pass Filter

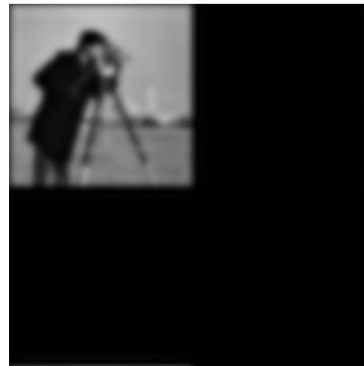
BLPF Output Image: Out



BLPF Centered Output Image: outc



BLPF outF1: Output Image with Zero Padding

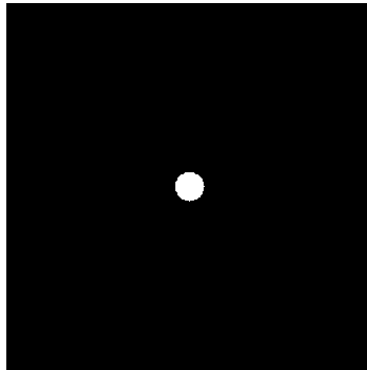


BLPF outF2: Output Image without Zero Padding

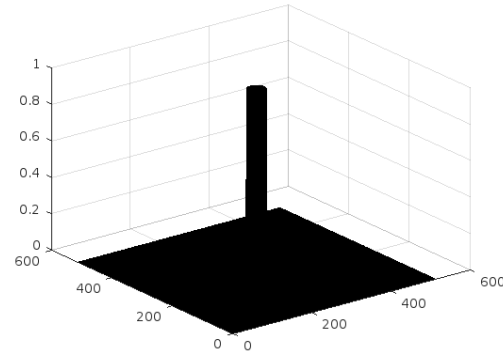


Ideal Low Pass Filter

2D view of H of Ideal LPF



3D View of H of Ideal LPF



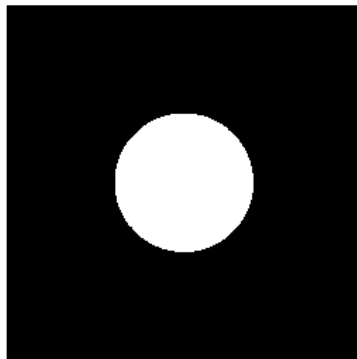
Ideal LPF outF2: Output Image without zero padding



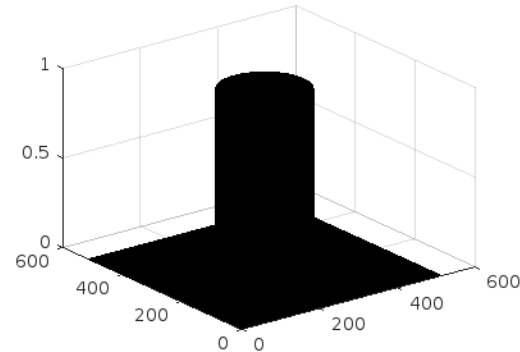
Original Image f



2D view of H of Ideal LPF $D_0 = 100$



3D View of H of Ideal LPF $D_0 = 100$



Ideal LPF outF2: Output Image without zero padding



Ideal Low Pass Filter

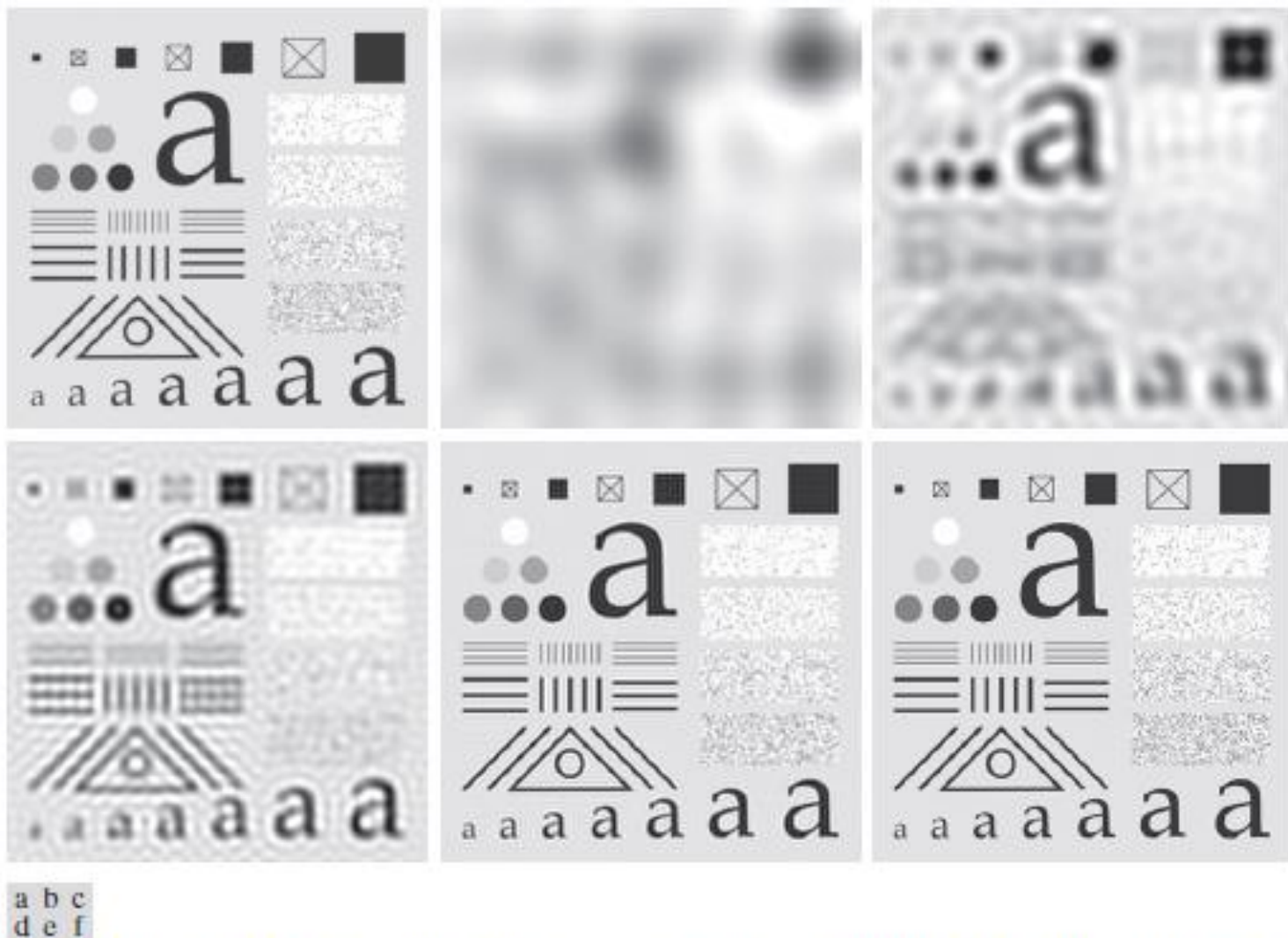


FIGURE 4.41 (a) Original image of size 688×688 pixels. (b)–(f) Results of filtering using ILPFs with cutoff frequencies set at radii values 10, 30, 60, 160, and 460, as shown in Fig. 4.40(b). The power removed by these filters was 13.1, 7.2, 4.9, 2.4, and 0.6% of the total, respectively. We used mirror padding to avoid the black borders characteristic of zero padding, as illustrated in Fig. 4.31(c).

Butterworth Low Pass Filter

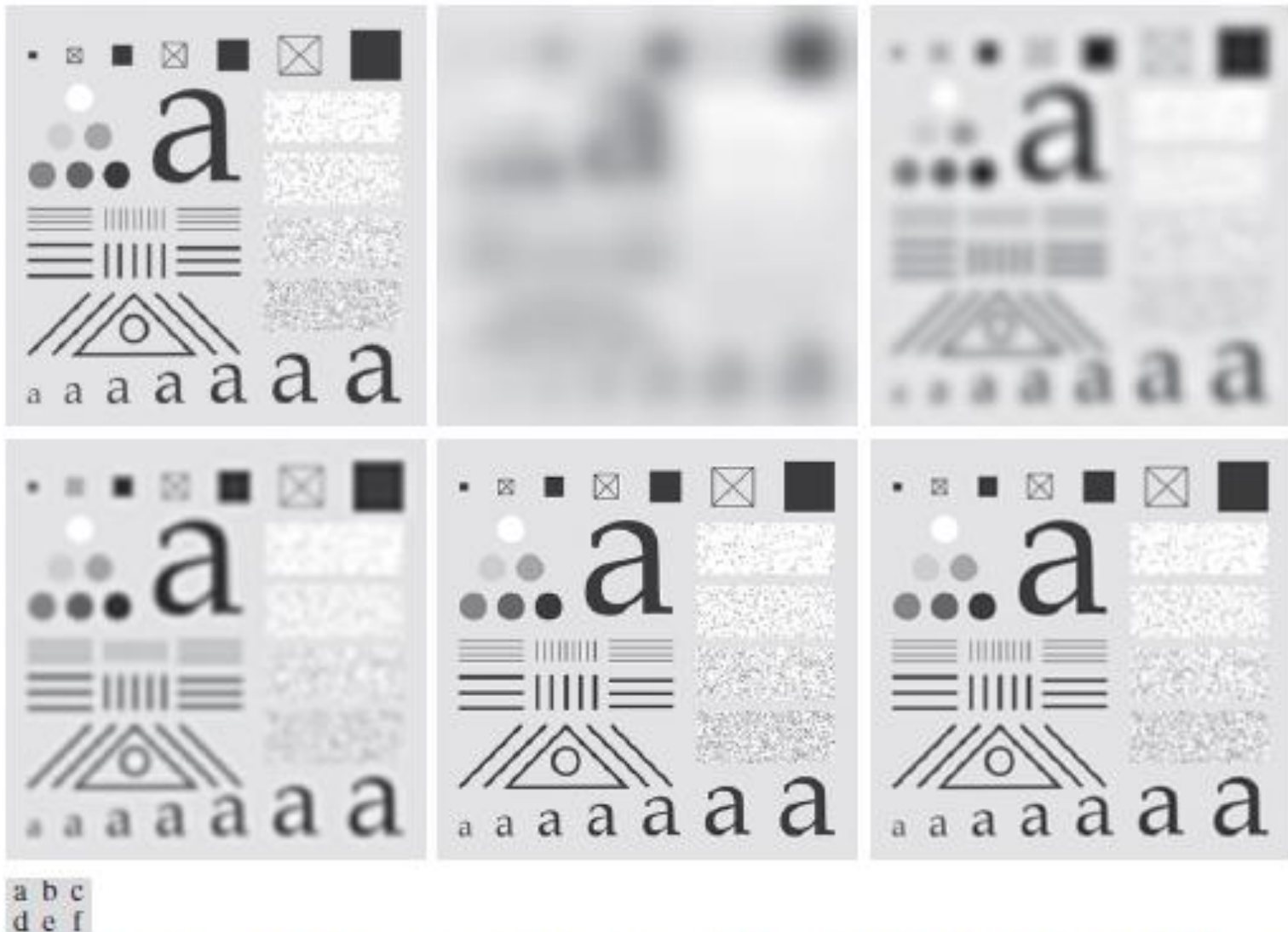


FIGURE 4.46 (a) Original image of size 688×688 pixels. (b)–(f) Results of filtering using BLPFs with cutoff frequencies at the radii shown in Fig. 4.40 and $n = 2.25$. Compare with Figs. 4.41 and 4.44. We used mirror padding to avoid the black borders characteristic of zero padding.

Gaussian Low Pass Filter

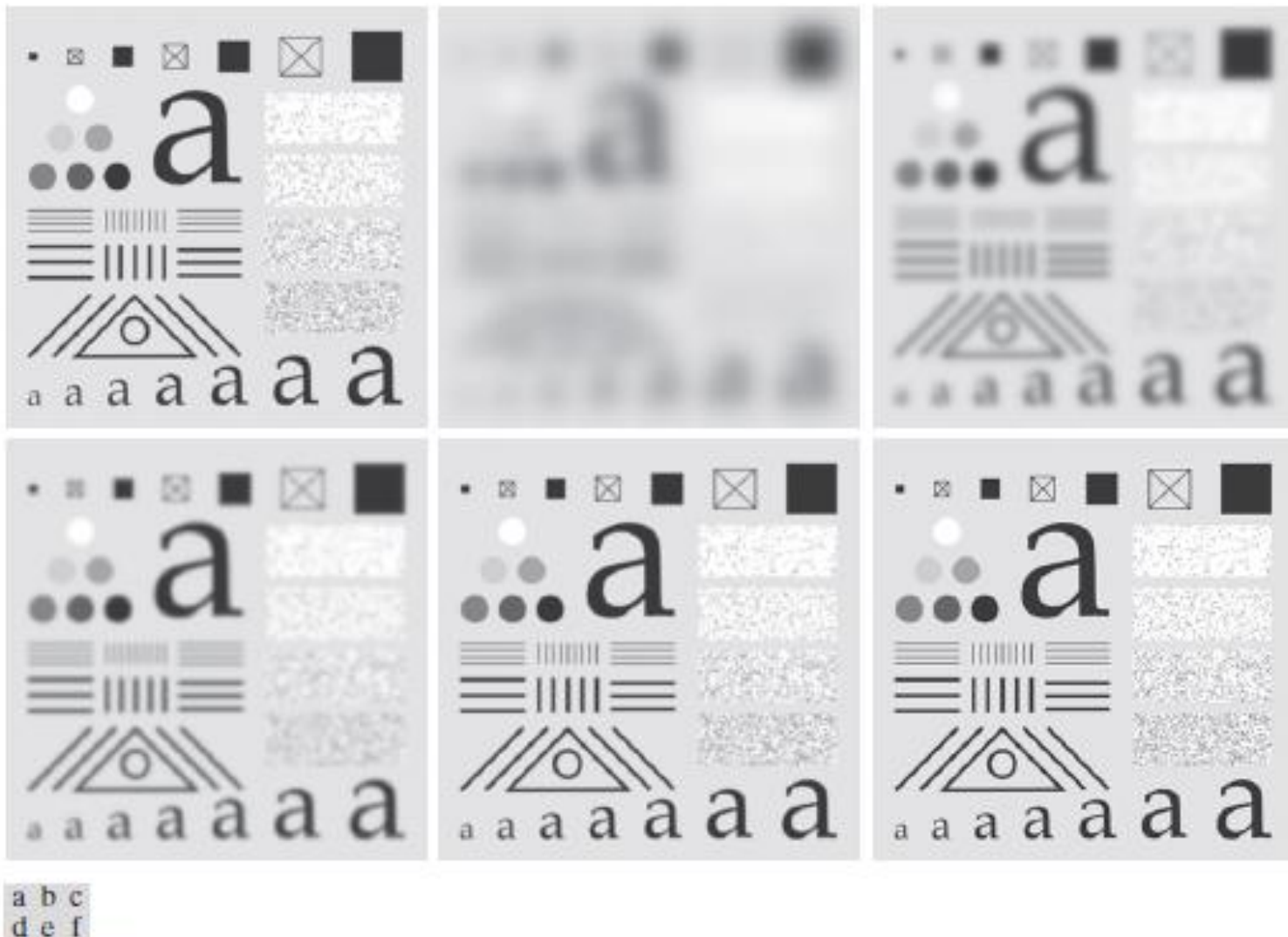


FIGURE 4.44 (a) Original image of size 688×688 pixels. (b)–(f) Results of filtering using GLPFs with cutoff frequencies at the radii shown in Fig. 4.40. Compare with Fig. 4.41. We used mirror padding to avoid the black borders characteristic of zero padding.

High Pass Filters

High Pass Filter

$$H_{HP}(u, v) = 1 - H_{LP}(u, v)$$

- It blocks the low frequency component and pass only high frequency component.
- **Edges & other abrupt changes** in gray levels are associated with high frequency components.
- **Image sharpening** can be achieved in the frequency domain by a high pass filtering process, which attenuates the low frequency components without disturbing high frequency components in the Fourier transform.

□ Types of High Pass Filter:

1. Ideal High Pass Filter (IHPF)
2. Butterworth High Pass Filter (BHPF)
3. Gaussian High Pass Filter (GHPF)

Ideal High Pass Filter

□ Mathematical Equation of Ideal High Pass Filter:

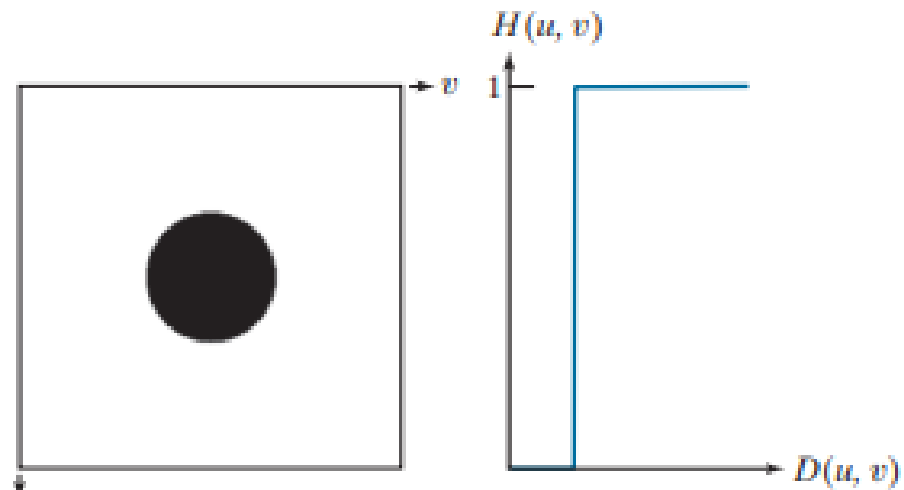
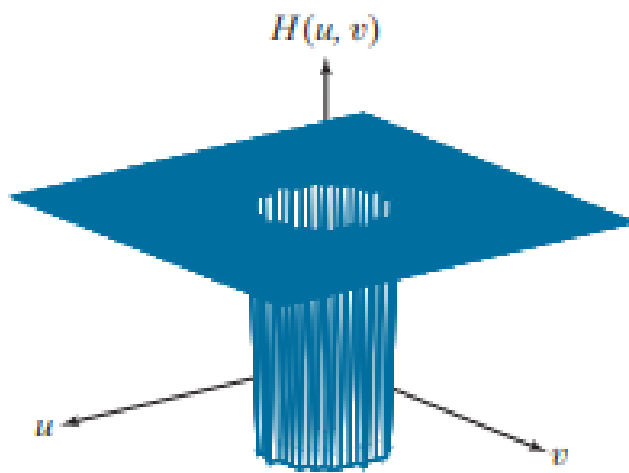
$$H(u, v) = \begin{cases} 0 & \text{if } D(u, v) \leq D_0 \\ 1 & \text{if } D(u, v) > D_0 \end{cases}$$

Where,

D_0 is a positive constant,

$D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the $P \times Q$ frequency rectangle; that is

$$D(u, v) = \left[\left(u - P/2 \right)^2 + \left(v - Q/2 \right)^2 \right]^{1/2}$$

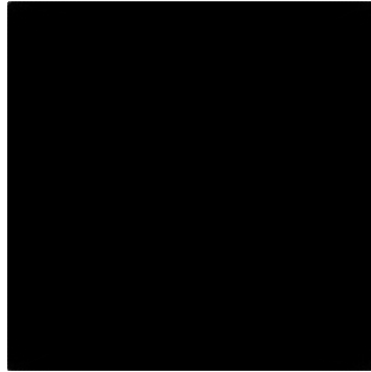


Ideal High Pass Filter

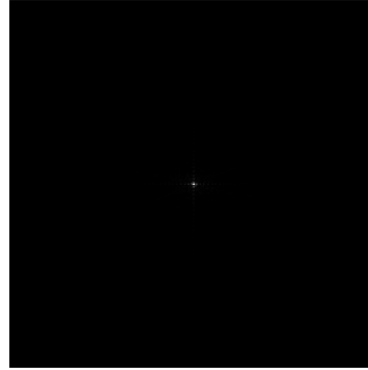
Original Image f



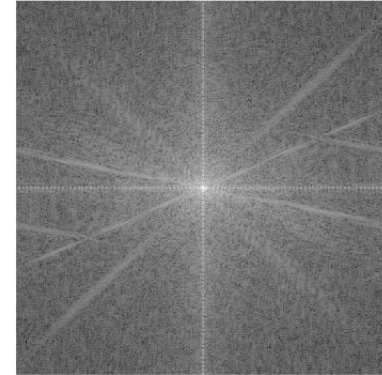
Fourier Spectrum F



Centered Fourier Spectrum F_c



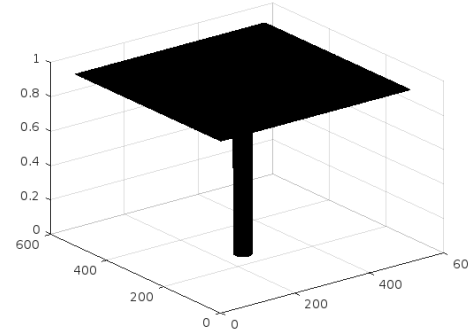
S_2 : Log Transformed F_c



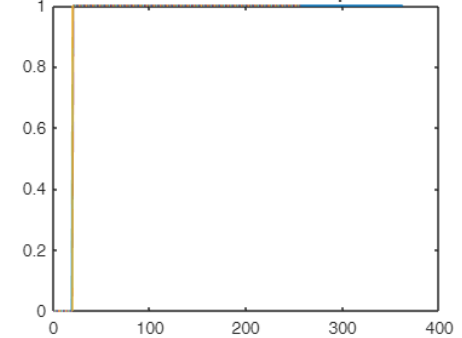
2D view of H of Ideal HPF



3D View of H of Ideal HPF

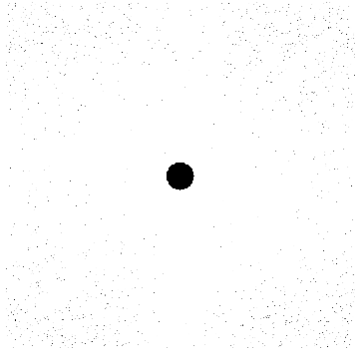


Transfer function of Ideal Low pass filter

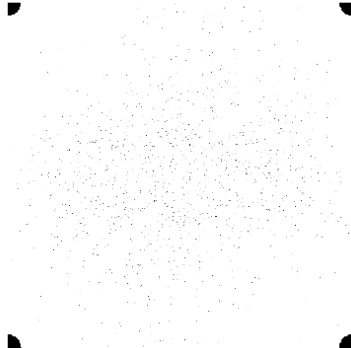


Ideal High Pass Filter

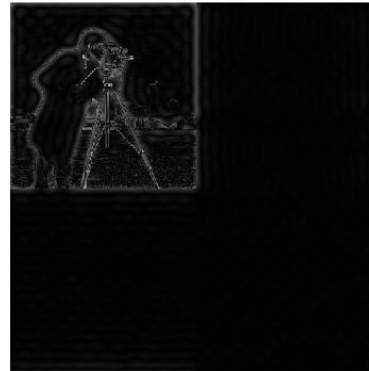
Ideal HPF Output Image: out



Ideal HPF Centered Output Image: out_c



Ideal HPF $outF1$: Output Image with Zero Padding



Ideal HPF $outF2$: Output Image without zero padding



Gaussian High Pass Filter

□ Mathematical Equation of Gaussian High Pass Filter:

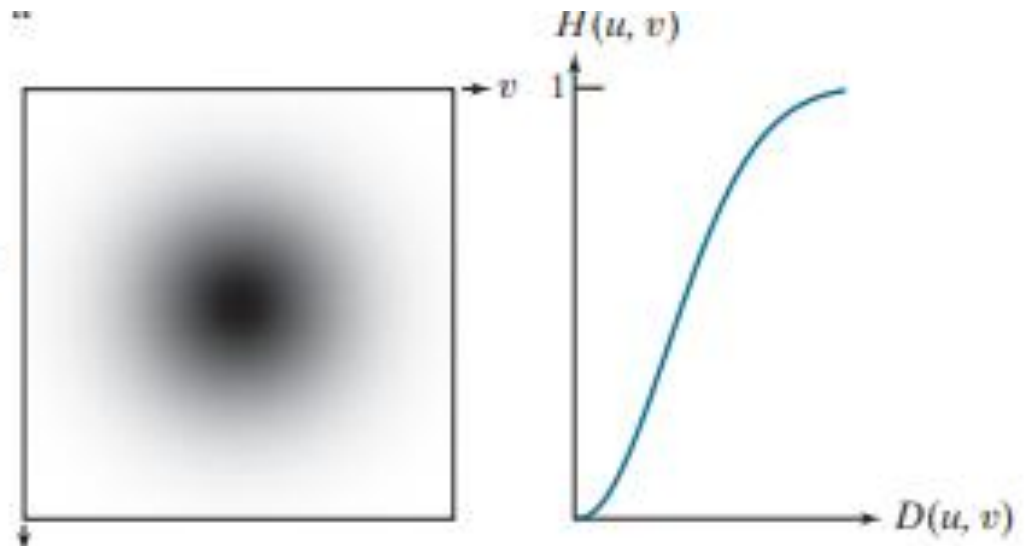
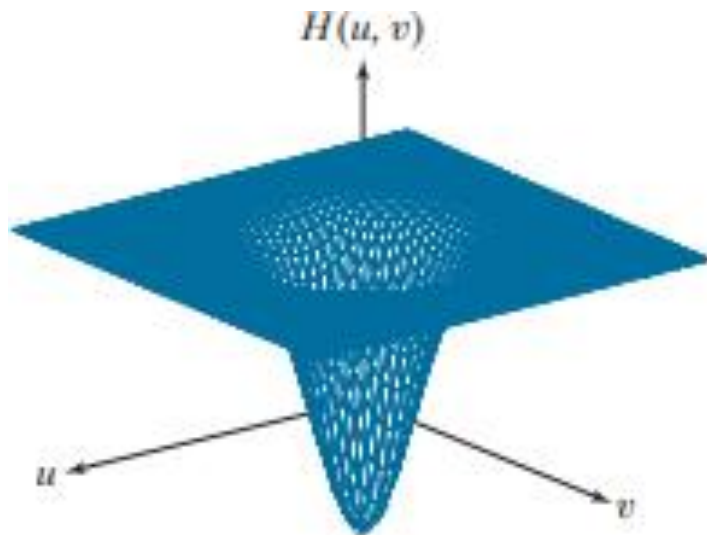
$$H(u, v) = 1 - e^{-D^2(u, v) / 2(D_0)^2}$$

Where,

D_0 : be the Cut-off frequency.

$D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the $P \times Q$ frequency rectangle; that is

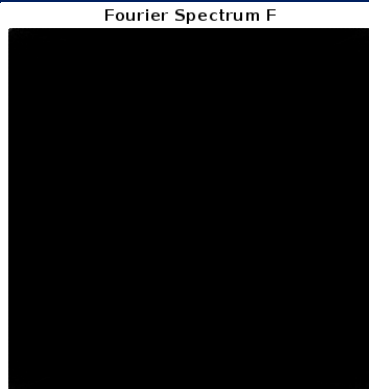
$$D(u, v) = \left[\left(u - P/2 \right)^2 + \left(v - Q/2 \right)^2 \right]^{1/2}$$



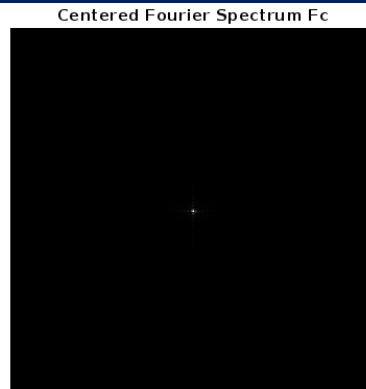
Gaussian High Pass Filter



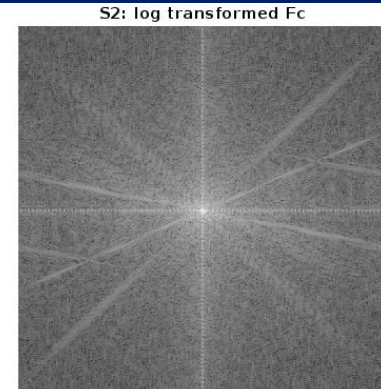
Original Image f



Fourier Spectrum F



Centered Fourier Spectrum F_c



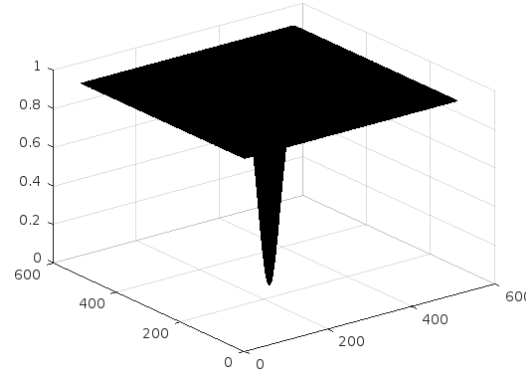
S_2 : log transformed F_c

Gaussian High Pass Filter

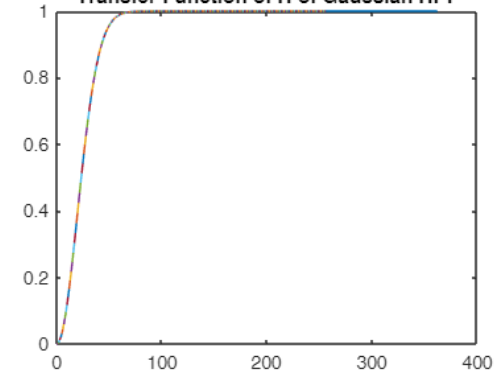
2D View of H of Gaussian HPF



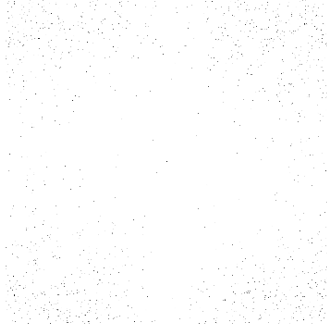
3D View of H of Gaussian HPF



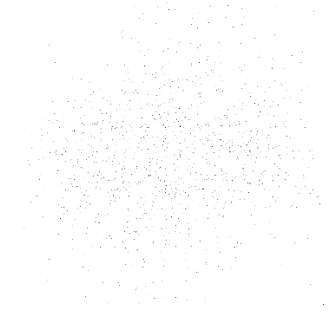
Transfer Function of H of Gaussian HPF



Gaussian HPF Output Image:out



Gaussian HPF Centered Output Image:outc



Gaussian HPF outF1:Output Image with Zero Padding



Gaussian HPF outF2:Output Image without Zero Padding



Butterworth High Pass Filter

□ Mathematical Equation of Butterworth High Pass Filter:

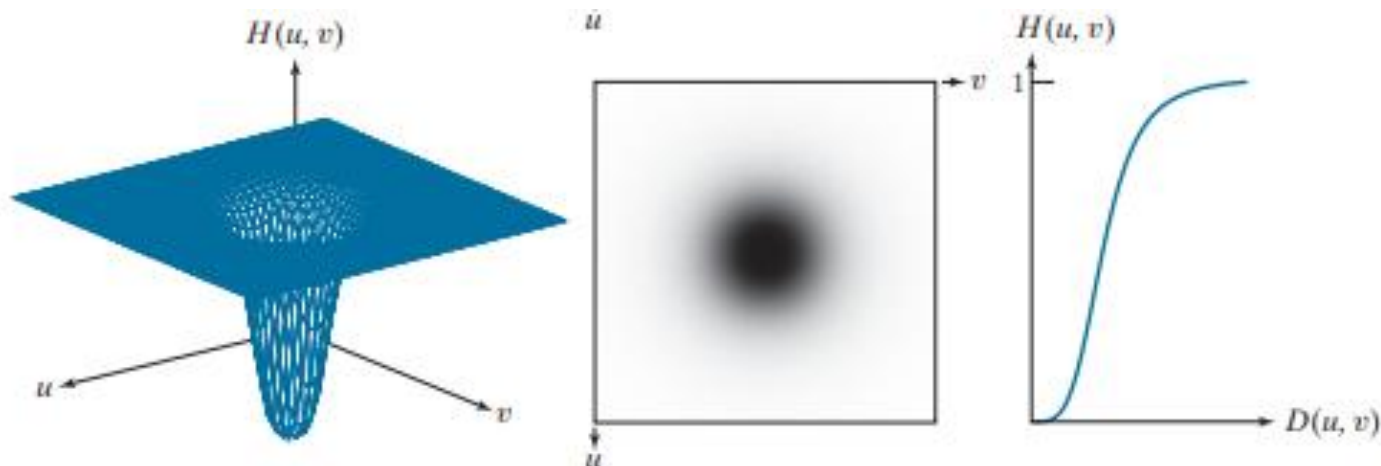
$$H(u, v) = \frac{1}{1 + \left[\frac{D_0}{D(u, v)} \right]^{2n}}$$

Where,

n : be the order of the filter and

$D(u, v)$ is the distance between a point (u, v) in the frequency domain and the center of the $P \times Q$ frequency rectangle; that is

$$D(u, v) = \left[\left(u - P/2 \right)^2 + \left(v - Q/2 \right)^2 \right]^{1/2}$$

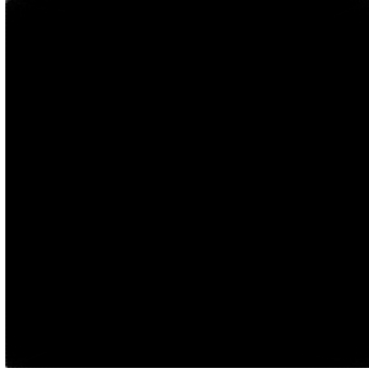


Butterworth High Pass Filter

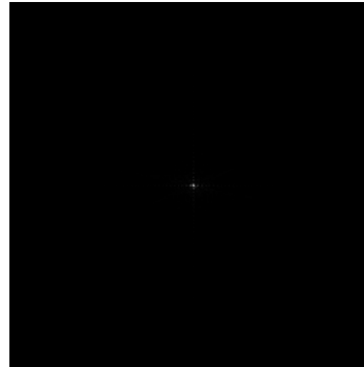
Original Image f



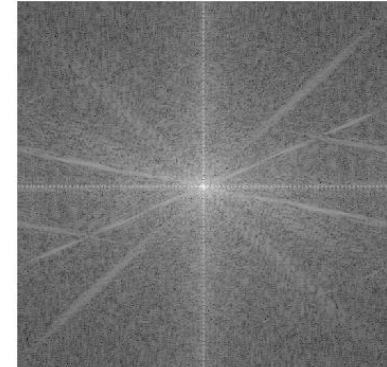
Fourier Spectrum F



Centered Fourier Spectrum F_c



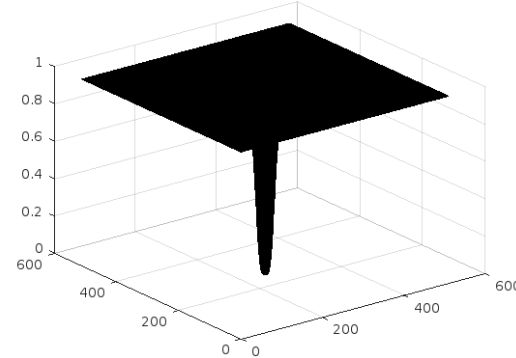
S_2 : Log Transformed F_c



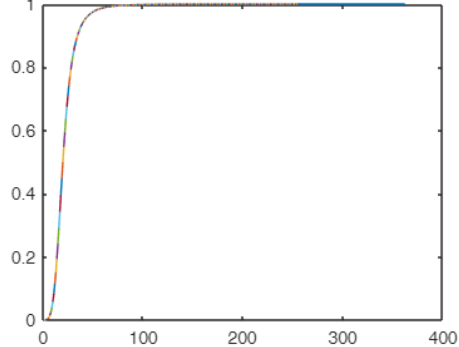
2D View of Butterworth HPF



3D view of H of Butterworth HPF

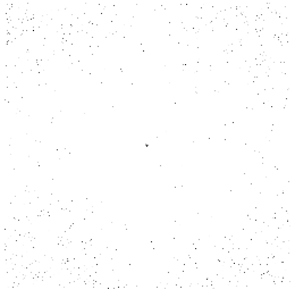


Transfer Function of H of Butterworth HPF

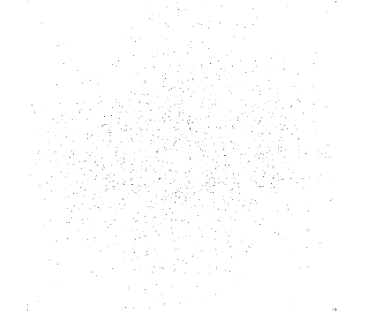


Butterworth High Pass Filters

Butterworth HPF Output Image: Out



Butterworth HPF Centered Output Image: outc



Butterworth HPF outF1: Output Image with Zero Padding



Butterworth HPF outF2: Output Image without Zero Padding

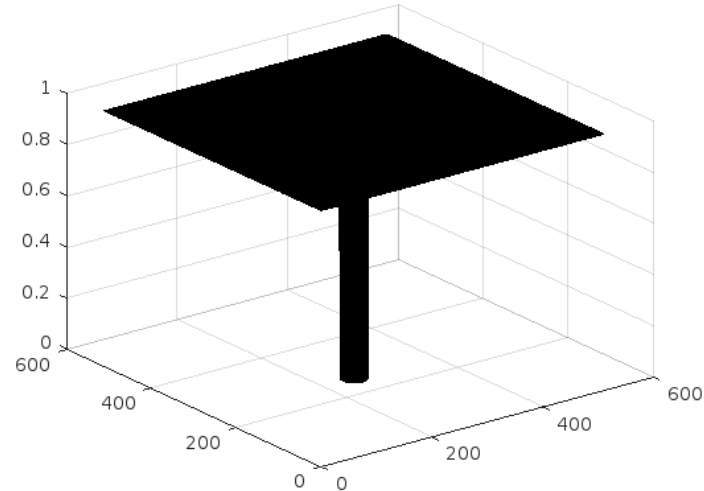


High Pass Filter

Original Image f



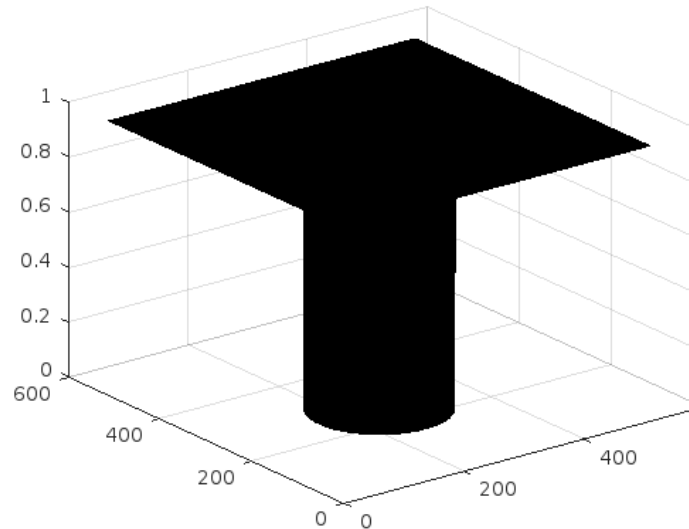
3D View of H of Ideal HPF



Ideal HPF outF2: Output Image without zero padding



3D View of H of Ideal HPF with $D_0=100$



Ideal HPF outF2: Output Image without zero padding $D_0=100$



High Pass Filter

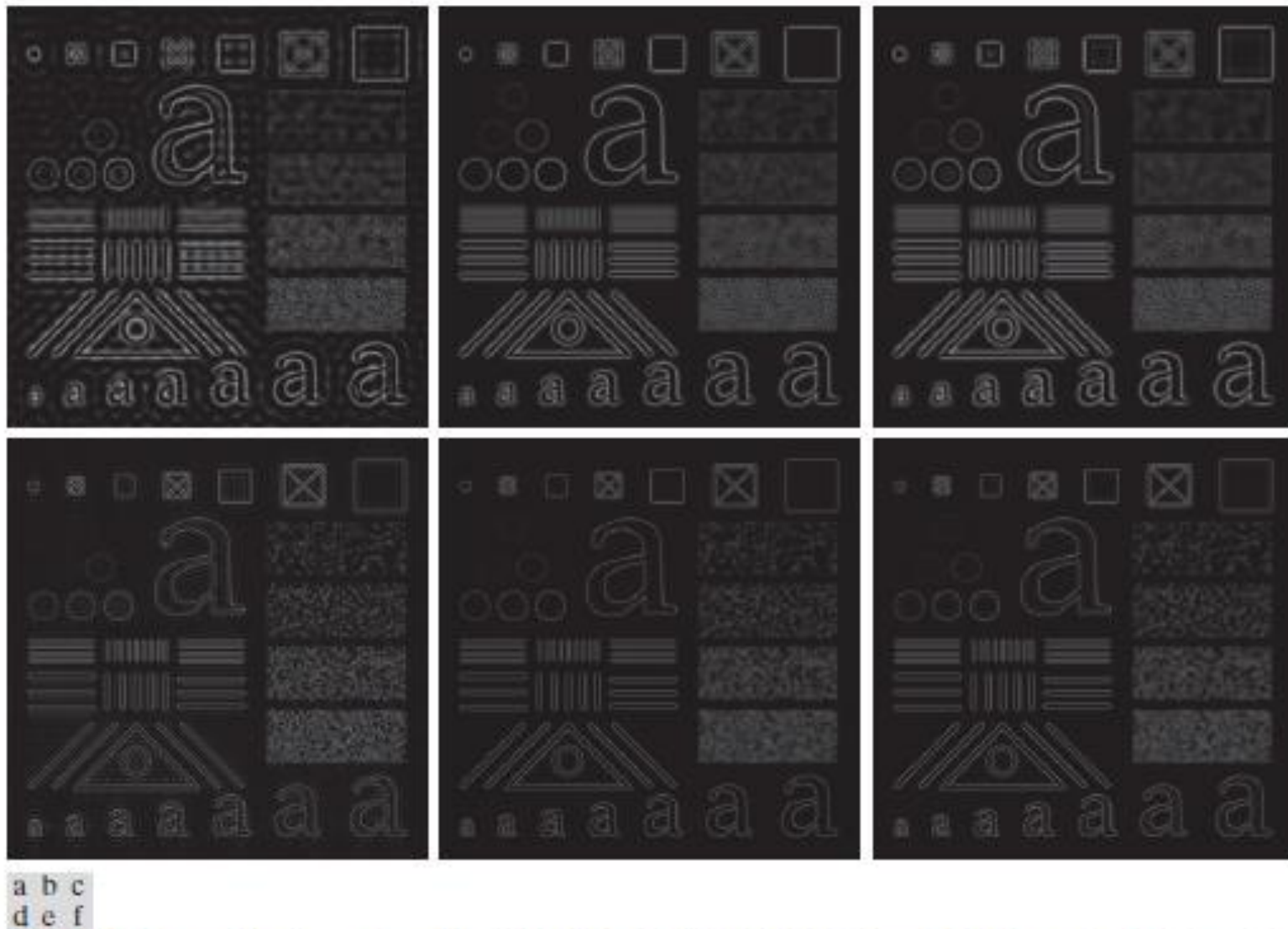


FIGURE 4.53 Top row: The image from Fig. 4.40(a) filtered with IHPF, GHPF, and BHPF transfer functions using $D_0 = 60$ in all cases ($n = 2$ for the BHPF). Second row: Same sequence, but using $D_0 = 160$.

University Questions

University Question

Ques 1: Compute row-wise Fourier Transformation for the following image, i.e. compute $F(0,v)$, $F(1,v)$, $F(2,v)$ and $F(3,v)$

GLA University, Mid Sem Examination, Even Semester, 2021-22

1	0	0	0
0	0	2	0
0	0	1	0
0	0	0	0

University Question

Ques2: Give the Formula to calculate 2D Discrete Fourier Transformation and its inverse. Find DFT of $\{0,1,0,0\}$.

GLA University, Mid Sem Examination, Odd Semester, 2022-23

University Question

Ques 3: The following image got corrupted while it was being captured. Find its Fourier Transform, So that it can be improved.

GLA University, Mid Sem Examination, Odd Semester, 2023-24

0	1	2	1
1	2	3	2
2	3	4	3
1	2	3	2

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

Twiddle Matrix

University Question

Ques 4: Compute the 2-D DFT for the signal shown in Fig. 7.

GLA University, Summer Carry Examination, Session, 2023-24

1	1	0	0
0	0	0	0
0	1	2	1
0	0	0	0

University Question

Ques 5: Compute the Fourier domain sequence $X(k)$ for the time –domain sequence $x(n)$ by applying DFT.

$$x(n) = \{1, 1, 0, 0\}$$

GLA University, End Sem Examination, Odd Semester, 2024-25



Thank You